Arav Palsule

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LinkedIn — GitHub

B.E. Data Science (2027) — CGPA: 8.34/10

Professional Summary

Aspiring Software Engineer with hands-on experience building web applications, backend workflows, REST API integrations, and data-driven software systems. Skilled in Python, JavaScript, TypeScript, SQL, React, and application development. Experienced in developing full-stack applications, integrating databases and external services, implementing application logic, and building automated processing pipelines. Strong foundation in problem-solving, debugging, and software development.

Education

A. P. Shah Institute of Technology, Thane 2023 – 2027
Bachelor of Engineering in Data Science **CGPA: 8.34/10**

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, SQL
Frontend Development: React, Vite, TypeScript
Backend Development: Flask, FastAPI, REST APIs, API Integration, Backend Workflow Development
Databases: MySQL, SQLite, Supabase
Software Development: Application Logic, Data Processing, Debugging, Testing, API Integration
Tools and Platforms: Git, GitHub, Zapier, Zoho
Additional Technologies: LangGraph, LLM Integration, FFmpeg, MoviePy

Projects

Radiator Routes – Full-Stack Travel Planning Platform
Developed a full-stack travel planning application using React, TypeScript, Vite, and Supabase, supporting trip creation, itinerary management, and personalized travel planning. Implemented application workflows for itinerary generation, travel memory, dynamic replanning, and integration with external travel data APIs. Built deterministic validation logic for budget constraints, scheduling conflicts, geographic feasibility, and opening hours, with automated LLM-assisted repair. Implemented group preference optimization using Least Misery scoring to compare itinerary alternatives.

CrackX – Road Condition Management System
Developed a road condition management application integrating road damage detection with image and video processing workflows. Integrated YOLO-based computer vision to detect and localize potholes and cracks, with bounding-box visualization and damage severity assessment. Connected detection workflows with road damage reporting and maintenance management functionality.

CineRec AI – Video Processing and Highlight Generator
Developed a Python-based video processing pipeline integrating scene analysis, emotion classification, scene ranking, and semantic music retrieval. Built processing workflows for timestamp generation, video clip extraction, soundtrack alignment, and automated reel composition. Integrated backend processing components using Python, FastAPI, FFmpeg, and MoviePy.

Dengue Outbreak Prediction – Data Processing Pipeline
Developed a Python data pipeline to process historical dengue cases, weather variables, geographic attributes, and temporal features for forecasting. Implemented preprocessing, feature engineering, and model training workflows using Pandas, NumPy, and regression algorithms.

Experience

AI Intern – Ennoble Social Innovation
September 2026 – October 2026

- Created custom Claude Skills using structured context engineering to automate business tasks, including PDF formatting according to internal document structures and standards.
- Developed an employee focus-monitoring prototype integrating face detection and emotion analysis to estimate focus scores and identify potential sleep or inactivity events.
- Designed screenshot capture and automated HR reporting workflows to communicate detected events and focus patterns.
- Built business process automations using Zapier, LLMs, and API integrations to streamline repetitive operational tasks.

Achievements

- **Rank 10** – C.R. Rao Statistics & Probability Olympiad: Ranked among 2.5 lakh+ participants across Asia.
- **AIR 3192, AICR 35** – Aryabhatta National Maths Competition 2026: Scored 100% in Round 1 and 80% in Round 2.
- **2nd Runner-Up** – QUASAR 4.0 Hackathon: Delivered a technical solution under a 24-hour development deadline.
- **Elite Rank** – NPTEL Multivariable Calculus: Recognized for mathematical and quantitative problem-solving skills.

Leadership

Founding President – Maths & Applied Science Club — 2026

- Founded and led the club, organizing 25+ technical events to promote mathematics, science, and technical learning among students.

Research Member – DSA Club — 2025 – 2026

- Explored software development, LLMs, RAG, and machine learning pipelines; mentored junior members in technical concepts.